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NLP Proposal Summary

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Named Entity Recognition for Medical Texts: Extracting Diseases, Treatments, and Symptoms from Clinical Reports

Project Overview

This project proposes an automated Named Entity Recognition (NER) system to identify and extract diseases, chemicals/drugs, and symptoms from medical reports. This "smart highlighting" capability provides superior accuracy on medical terminology and serves as a foundation for automated clinical coding and drug discovery research.

Objectives

- Prepare and preprocess biomedical datasets for medical entity recognition tasks.
- Develop a high-performing model to accurately classify medical entities.
- Create an interactive demonstration tool for real-time entity visualization.

Methodology

1. **Data Acquisition:** Utilize NCBI Disease Corpus and BC5CDR dataset, providing thousands of PubMed abstracts with entity tags for diseases, chemicals, and symptoms.
2. **Preprocessing:** Clean and tokenize text using WordPiece tokenizer, aligning labels to tokens in IOB tagging format to handle multi-word entities.
3. **Model Architecture:** Fine-tune BioBERT with an added token-classification layer (linear layer with softmax) to predict entity tags for each token.
4. **Training:** Optimize the model using cross-entropy loss with appropriate regularization and validation monitoring to ensure generalization.
5. **Visualization:** Implement SpaCy for real-time entity rendering with color-coded tags.

Tools and Technologies

- **Programming & Frameworks:** Python 3.9+ with PyTorch/TensorFlow, Hugging Face Transformers (BioBERT), SpaCy (visualization), and Scikit-learn (evaluation).
- **Development Environment:** Google Colab with GPU acceleration for efficient training, Git/GitHub for version control and collaborative development.

Expected Outcomes

- A fine-tuned NER model processing medical text and returning structured entity data
- Performance report with precision, recall, F1-score metrics and error analysis.
- Interactive Python notebook with color-coded entity visualization.
- Final documentation detailing methodology, results, and healthcare applications.

